Figure 1: Conveyor system with adjustable guide rails and photoelectric sensors for position mapping and trigger synchronization.

Figure 1: Lab setup with Mitsubishi RV-8CRL-D robot, encoder-based conveyor tracking system, and CR800-D controller.

Expt No:
05
Date:
05 / 03 /2026

DETERMINISTIC CALIBRATION AND ENCODER SYNCHRONIZATION

1. **AIM:**

   To perform high-precision coordinate calibration between a linear transport system and a 6-axis robotic manipulator by mathematically mapping rotational encoder pulses to linear spatial displacement (P_EncDlt) on the Mitsubishi RV-8CRL-D industrial system.

2. **APPARATUS REQUIRED:**
   a) Manipulator System: Mitsubishi RV-8CRL-D 6-axis industrial robot.
   b) Control Hardware: CR800-D series dedicated robot controller.
   c) Software Suite: RT ToolBox3 integrated engineering environment.
   d) End-of-Arm Tooling : Precision drawing effector configured within the tool coordinate system.
   e) Infrastructure: Solid installation platform and standard TCP/IP or standard USB communication interface.

3. **THEORY:**

   Trajectory generation for planar symbolic mapping requires the mathematical decomposition of characters into primitive geometric segments.
   a) Core Principle:

      For an industrial robot to accurately interact with moving objects on a conveyor (Dynamic Conveyor Tracking), the controller must mathematically correlate the physical displacement of the conveyor belt with the Cartesian coordinate system of the manipulator. This requires establishing a precise pulse-to-millimeter ratio.
   b) Working Mechanism:
      i) Pulse Accumulation: As the conveyor moves, a connected rotary encoder generates digital pulses. The controller continuously reads this accumulated value via the M_ENC system variable.
      ii) Spatial Reference Mapping: By physically aligning the robot with a target on the conveyor at two distinct locations (upstream and downstream) and recording both the absolute robot coordinates (P_CURR) and the encoder pulse counts at those exact moments, a mathematical slope vector is derived.
      iii) Overflow Correction: Incremental encoders have a maximum logical threshold before the pulse count resets or rolls over. The algorithm must account for boundary crossings (e.g., the $\pm80,000,000$ pulse threshold) to maintain tracking integrity.
   c) Key Formula:

      The linear spatial displacement per single encoder pulse (P_EncDlt) across any axis is calculated as:

$$P_{EncDlt} = \frac{P_2 - P_1}{EncDiff}$$

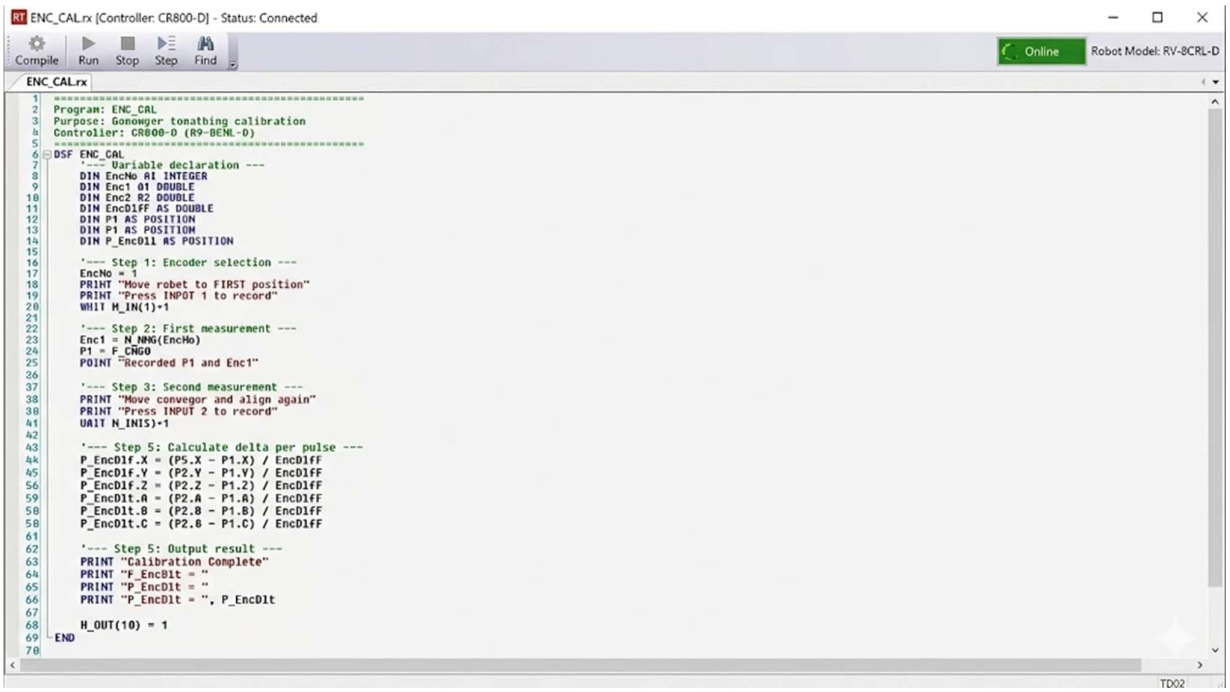

Figure : RT ToolBox3 Source Code for Conveyor Tracking Calibration.

4. PROCEDURE:

a) Setup & Connections:

Interface the conveyor's rotary encoder signal to the CR800-D controller's tracking card. Ensure the external trigger inputs (Input 1 and Input 2) are mapped to the teaching pendant or physical pushbuttons to capture the measurement snapshots.

b) Algorithm & Logic:

i) Initialize variables and select the target encoder channel (EncNo = 1).

ii) First Measurement: Jog the robot to a physical reference point on the upstream section of the conveyor. Trigger Input 1 to record the current position (P1) and pulse count (Enc1).

iii) Second Measurement: Run the conveyor to move the reference point downstream. Jog the robot to perfectly realign with the reference point. Trigger Input 2 to record the new position (P2) and pulse count (Enc2).

iv) Delta Calculation: Calculate the raw pulse difference (EncDiff) and apply the necessary overflow/underflow correction logic.

v) Compute the coordinate delta (P_EncDlt) for all 6 degrees of freedom (X, Y, Z, A, B, C).

c) Execution Steps:

i) Execute the ENC_CAL program in manual/step mode.

ii) Follow the generated PRINT prompts on the teaching pendant to align the robot TCP with the physical target and trigger the respective inputs.

iii) Verify the final calculated P_EncDlt matrix outputted on the pendant screen.

d) Observation Instructions:

Analyze the resulting coordinate delta values. Assuming the conveyor moves strictly along one primary Cartesian axis (e.g., the robot's Y-axis), the calculated delta for that specific axis should be a distinct non-zero value representing the millimeter-per-pulse ratio, while the remaining orthogonal axes (X, Z) and rotational postures (A, B, C) should theoretically calculate to zero.

5. RESULT AND OBSERVATION:

The system successfully derived the pulse-to-metric conversion factor P_{EncDlt}. The deterministic relationship established during calibration ensures that subsequent conveyor tracking operations maintain high spatial accuracy, as verified by functional evaluations.